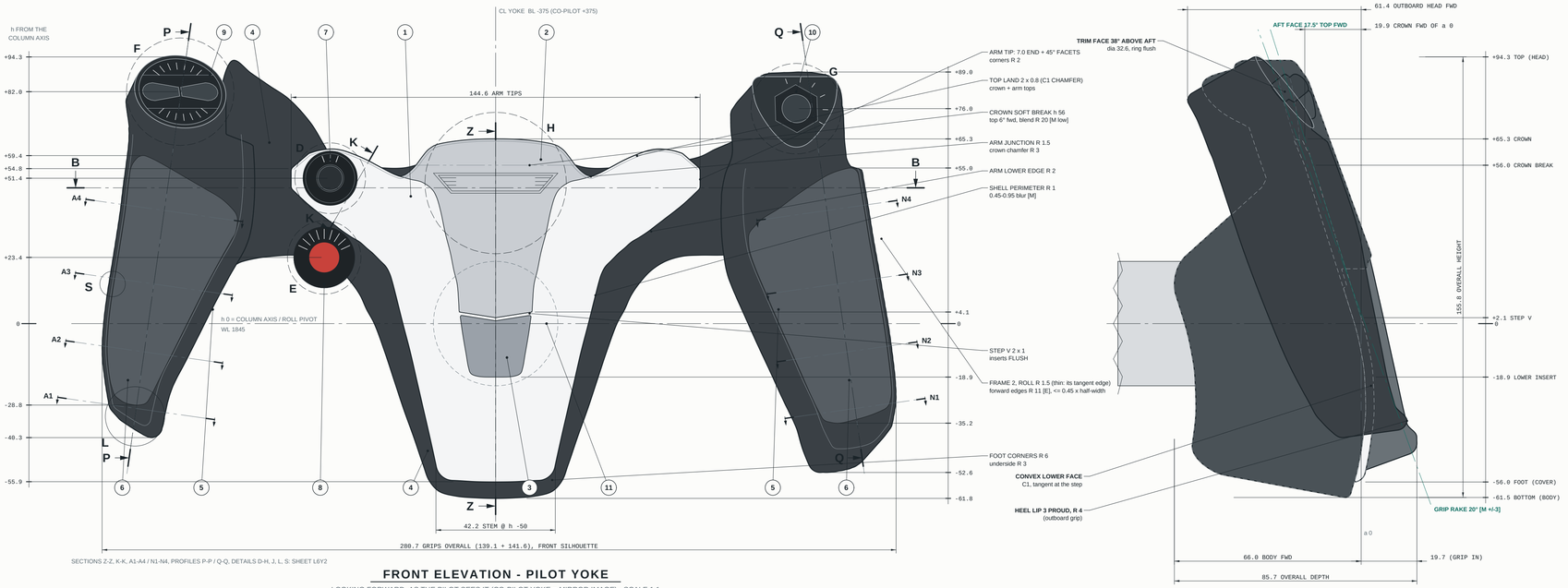


SECTIONS Z-Z, K-K, A1-A4 / N1-N4, PROFILES P-P / Q-Q, DETAILS D-H, J, L, S: SHEET L6Y2

FRONT ELEVATION - PILOT YOKE

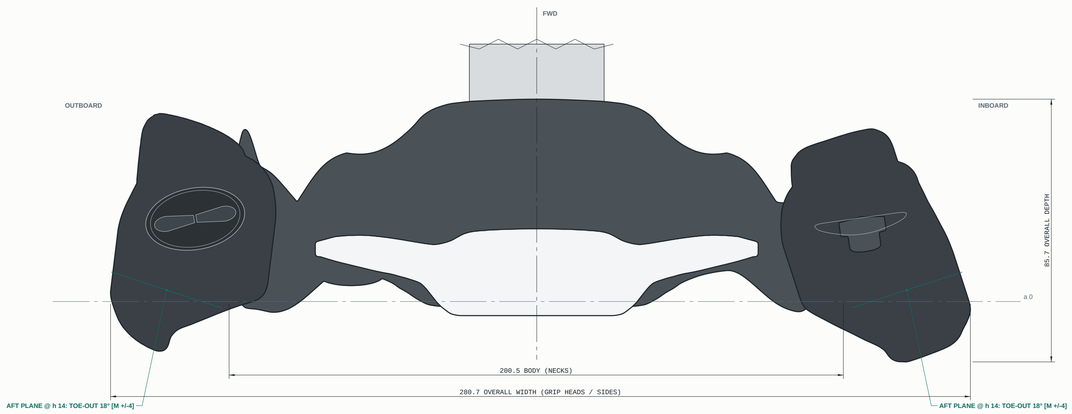
LOOKING FORWARD, AS THE PILOT SEES IT (CO-PILOT YOKE = MIRROR IMAGE) - SCALE 1:1

SIDE ELEVATION

FROM OUTBOARD, NOSE LEFT, NEUTRAL PITCH; a 0 = X 3700; COLUMN BROKEN (L6Y2 DETAIL J) - 1:1

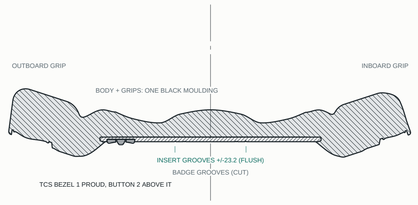
GRIPS PROVISIONAL: RAKE / TOE / a0 FITTED TO 5 VIEWS [M +/-3° / +/-4° / +/-3-5]; THE AOPA PHOTO DISAGREES (OPEN POINT 2)

BODY (HIDDEN): TOP 42° FROM THE HORIZONTAL, TOP-TO-FRONT R 6, DEPTH 60 [E], FOOT R 14 [E] (L6Y2 Z-Z)



VIEW FROM ABOVE

FORWARD UP, PILOT YOKE - SCALE 1:1; GRIPS' FORE-AFT LIE PROVISIONAL (OPEN POINT 2, L6Y2 PHOTO CHECK)



SECTION B-B

h 48 (BADGE / TCS), EXACT CUT OF THE SOLIDS, FROM ABOVE - SCALE 1:2

ITEMS & FINISHES (render/lookdev names; 'new' = to add)

#	PART	MATERIAL	FINISH / COLOUR [SOURCE]
1	WHITE COVER (Y SHELL)	yoke_white	gloss clear-coat white, neutral-cool (0.77 / 0.78 / 0.80), fine pearl; R 1 edges, top land [M]
2	UPPER INSERT + CROWN, BADGE	yoke_silver (new)	satin aluminium 0.62, metallic 0.85, rough 0.30; flush, 0.6 x 0.5 grooves [M]
3	LOWER INSERT	yoke_silver_dark (new)	smoke silver 0.30, metallic 0.85, rough 0.34; flush [M]
4	BLACK BODY (RIM, WEB, NECK)	grip_black	satin black, rough 0.42: crisp edge highlights; R 2 aft edge [M]
5	GRIP (2) WITH FRAME, HEEL LIP	grip_black	satin black; 2.0 frame R 1.5, heel lip 3 proud R 4 [M]
6	LEATHER PAD (2)	leather_black	charcoal 0.05, rough 0.58, 1.1 pebble grain, ~1 creases, 2.5 below the frame, groove 1.0 x 0.5 + bead 0.4 at its edge [M / E]
7	TCS BUTTON	gloss_black / black	gloss black bezel 1 proud [E], satin domed button 2 proud of it [M low]
8	AFCS DISCONNECT	button_red (new) / black	red cap 0.70 / 0.035 / 0.04, clear coat 0.3, NOT emissive, on a black boss [M]
9	PITCH TRIM ON ROUND FACE	black / legend_white	satin black face, ring + rocker, tick legends 0.80 [M]
10	PUSH BUTTON ON SHIELD	black	satin black hex collar, smooth cap, centre dimple [M]
11	COLUMN	metal_dark	satin dark metal, dia 44 [E] (never seen)
12	COLUMN SURROUND (PANEL)	pedestal_gunmetal	satin dark grey J band 23 wide + dished recess [M / E]

MAIN DIMENSIONS (read from the table / the solids)

DIMENSION (mm, YOKE-LOCAL)	VALUE	SOURCE
shell height / about h 0	121.3 / -56.0..+65.3	[M] rev J landmark-pose frame
arm tips / stem at h 0 / at h -50	144.6 / 65.6 / 42.2	[M] sigma 1.5
crown above the arm junctions	13.3	[M] parting grooves
step / lower insert bottom	h 2.1 (V) / -18.9	[M]
black rim outboard of the stem at h 10 / 0 / -50 / under the foot	12.6 / 10.1 / 4.5 / 5.8	[M] body - shell outlines
aft face above the step	17.5° top fwd	[M] bundle adj., +/-4
body depth / top / cover	60 / 42° from horiz. / 3	[E] / [M lower bound] / [E]
heads top: FRONT SILHOUETTE out / in	94.3 / 89.0	[M] solids
heads top: AFT-FACE OUTLINE out / in	97.4 / 90.0	[M] table
grip bottoms (front silhouette) out / in	-40.3 / -52.6	[M]
grips overall: front silhouette (out + in)	280.7 (139.1 + 141.6)	[M] solids
grips overall: aft-face outlines (out + in)	281.3 (139.3 + 142.0)	[M] table
rake / toe-out / a0 @ h 14: out; in	20° / 18° / -3.75; 20° / 18° / -3.75	[M] 5-view joint fit, +/-3-5
walls (0 = aft normal, 1 = fore-aft) out / in	0.6 / 0	[M] face-on silhouettes
grip depth (normal) bottom / mid / head, out; in	20 / 27 / 30; 22 / 27 / 30	[M] / [M] / [E], PCHP
pad out: top / slot corner / bottom	59.4 / (91, 38) / -28.8	[M] diagonal top 31°
pad in: top / slot corner / bottom	55.0 / (89.4, 32) / -35.2	[M] diagonal top 33°
frame / leather below it / heel lip	2 / 2.5 / 3 proud R 4	[M]
trim face centre / dia / up, turn	(111.5, 82) / 32.6 / 38°, 10°	[M] triangulated, 7 views
shield facet / button centre	31 x 25.5 / (106.3, 76)	[M] (inboard: -u)
TCS / AFCS cap / boss	(58.5, 51.4) 19.2 / 10.8 x 5.5 / 21.6	[M]
column dia / axis	44 [E] / h 0 = WL 1845	kept; surround reads 1821
pitch travel / roll	+/-90 [E] / +/-70 [E]	unchanged (YOKE)

KNEES vs YOKE (L6B), REV H > I > J > K - CRITERIA UNCHANGED

OCCUPANT	LEVEL mm	ROLL+/-70 mm	CONTACT deg	KEEP 25 deg	RUDDER mm
95th male	38 > 35 > 40 > 37	-19 > -18 > -19 > -19	17.5 > 17.5 > 15.0 > 15.0	5.0 > 2.5 > 5.0 > 5.0	-11 > -13 > -9 > -11
50th male	50 > 48 > 52 > 50	-13 > -14 > -14 > -14	25.0 > 25.0 > 22.5 > 22.5	10.0 > 10.0 > 10.0 > 10.0	5 > 3 > 7 > 5
5th male	51 > 48 > 51 > 48	-21 > -21 > -22 > -22	22.5 > 22.5 > 22.5 > 20.0	10.0 > 10.0 > 10.0 > 10.0	20 > 18 > 22 > 20
5th female	76 > 78 > 81 > 79	5 > 0 > 0 > 0	87.5 > 87.5 > 87.5 > 87.5	27.5 > 25.0 > 25.0 > 22.5	55 > 53 > 56 > 53

NOTES

1. Yoke-local mm: u outboard (pilot: -y), h up from the hub centre = the column axis / roll pivot (WL 1845, BL -/+375), a aft of the aft face at the insert step (X 3700 at neutral pitch). Drawn from model/interior.py YOKE_DESIGN rev K through the solids of model/yoke_geom.py; L6 / L6B take the same solids (yoke_outline_yz, yoke_sweep_pieces).
2. CURVE LAW (table curve_law): every outline is a centripetal Catmull-Rom spline through its samples, sampled every 0.25; the samples in the *_corners tables are corners (R 0 sharp, R > 0 a tangent fillet); the lower aft face is a C1 Hermite tangent to the 17.5° plane. The solids use the EXACT distance to those polylines (no raster); views are exact silhouettes, sections exact cuts.
3. STAGE-3 SURFACE ROUTE: patches, not an implicit mesher -- each face (cover face and inserts, grip aft plane with its pad pocket, body faces) a grid_surface over its analytic parametrisation trimmed by cad.mesh.trim with these exact 2-D fields; each tabled edge round a sweep_profile of its radius along the outline; creases only where tabled; vertex normals from the analytic surfaces. The SDF solids stay the inside / outside oracle for fit_check and these sheets. Acceptance (model/yoke_checks.py, L6Y2): vertex normals within 2° of the analytic at the 95th percentile, no undetailed section coups, nothing above the trim face (shield.
4. White moulding symmetric; the black body and the grips are NOT: an outboard web up to the trim head, an inboard LOW NECK with an open notch under the inboard head; switches outboard (TCS, AFCS, trim), the push button on the inboard head; the co-pilot yoke is the mirror image.
5. NO MARKINGS: the badge is an abstract winged plate, every legend a row of ticks.
6. Not modelled / unsupported: the rev H column bellows (none in 8 photos), a PTT trigger (none seen), the PC-24's extra grip tab.
7. GRIPS' 3-D LIE: rake / toe / a0 / walls fitted jointly per grip to the curated silhouettes and pad traces of 7 views (L6Y2 photo check): the 5 views from inboard (23-27° off aft, Mexico 15-20° from above) land within 0.7° / 1.4° / 1.2 mm of the table (rake +/-3° out / +/-5° in, a0 +/-3 / +/-5 for a 10 % cost rise); the AOPA photo (one camera) disagrees by 5-15 mm and no plane the others allow meets it -- open point 2.
8. Knee checks on the rev K solids: the 95th clears the yoke by 37 mm wings level (criterion 25: passes); roll to contact 15.0° (rev J 15.0°), keeping 25 mm to 5.0° -- owner decision (open point 1 of L6B; criteria not loosened).

OPEN POINTS FOR THE OWNER

1. Absolute fore-aft station: the locked photos put the aft face 60 +/-30 forward of the drawn neutral (hub_dx kept: the gust-lock pose?).
2. Grips' fore-aft lie: the AOPA photo wants the aft faces 5-15 further aft below h 0 (and puts the inboard button 13 off); 5 other views agree with the table -- a true side photo of a PRO / PC-24 yoke would settle it.
3. Column: the surround's joint line reads WL 1821 +/-10 (25 under the pivot, kept); J band or a full collar round the column (L6Y2 detail J); push travel 90 [E].
4. Shell skew: posed on the symmetric landmarks the stem sits ~1.6 inboard and the arm tips ~1.5 outboard of the inserts' centreline in all 7 views (TCS / AFCS u fit within 0.7): carry the <= 2 mm skew or keep it symmetric (kept).
5. Knees: the 95th meets a grip from 15.0° of roll (rev I 17.5°): owner decision, criteria unchanged.
6. Crown soft break (R 20 [E]) vs a crease; hidden: body depth 60, the necks' depth, rocker / cap heights.

REVISIONS - PC12-L6Y

REV	DATE	DESCRIPTION
A	2026-10-05	first issue (owner 2026-10-05: 'the yoke isn't quite right especially if you look at the details'): yoke re-measured on 8 PRO photos; YOKE_DESIGN rev I replaces the rev H outline
B	2026-10-05	rev J (judges r1): the yoke as solids (model/yoke_geom.py) -- every view / section exact; curve law + corner tables; one body with an outboard web and an inboard LOW NECK (open notch under the head); pads with diagonal tops in a raised 2 mm frame, rolled heel lip; grips refitted per grip (toe 18 / 18) on 4 pad views; ROUND trim face and SHIELD facet heads, triangulated switch centres; crown / arm tops 1.7 lower (landmark-pose frame), top land, crown crease; body top falling forward 35 deg; inserts flush; AFCS round boss, hex button collar, mirrored rocker lobes; sheet L6Y2 added (sections, heads, switches, column, photo check)
C	2026-10-05	rev K (judges r2): exact 2-D distance fields (no raster), Stage-3 surface route + acceptance checks; grip walls fore-aft (outboard 0.6) -- slot-side and heel silhouettes on the curated photo data; forward-edge R capped, PCHIP depth, heel lip R 4, pad groove + bead built; body top 42 deg from horizontal, top-to-front R 8; the web held under the trim face; AFCS boss, TCS, rocker, hex button as solids; TCS button proud; crown soft break R 20; grip planes JOINT-FITTED to 7 views (5 confirm the table +/-3 deg / +/-3-5 mm; the AOPA photo disagrees: open); outboard heel corner pulled in; ordinate dimensions, cutting planes, detail circles, all values computed

PILATUS PC-12 PRO

PC-24-STYLE CONTROL YOKE - VIEWS

FIRST ANGLE PROJECTION

DWG NO	REV	SCALE	SHEET	UNITS
PC12-L6Y	C	AS SHOWN	A1 · 1 OF 2	mm

DRAWN: Claude - loftkit parametric model

DATE: 2026-10-05

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION